

DATADOSE CLINICAL DECISION INTELLIGENCE PLATFORM

DEPI x eYouth 2026 Graduation Project



Data Dose



Table of Contents

1. **Project Overview**
2. **Problem Statement**
3. **Solution Overview**
4. **Project Objectives**
5. **System Architecture**
6. **Neo4j Knowledge Graph**
7. **Data Pipeline**
8. **Real-time Data Streaming**
9. **Analytics Warehouse**
10. **Airflow Orchestration**
11. **AI Clinical Features**
12. **Application API**
13. **Technology Stack**
14. **Power BI Reporting**
15. **Docker Deployment**
16. **Key Performance Indicators (KPIs)**
17. **Metrics and Monitoring**
18. **Team and Roles**
19. **Challenges and Lessons Learned**
20. **Conclusion**

1. Project Overview

DataDose is a cutting-edge Clinical Decision Intelligence (CDI) platform designed to empower healthcare professionals with real-time, actionable insights. Developed as the graduation project for DEPI x eYouth 2026, DataDose aims to revolutionize patient care by leveraging advanced data analytics, artificial intelligence, and a robust knowledge graph.

This platform integrates diverse data sources, processes them through a sophisticated data pipeline, and presents critical information through an intuitive interface and comprehensive reports. The project

emphasizes a scalable and reliable architecture, facilitating informed decision-making at the point of care.

2. Problem Statement

The healthcare industry faces significant challenges in managing and interpreting the vast amount of patient data generated daily. Clinicians often struggle with:

- **Data Silos:** Patient information is frequently fragmented across various systems, making it difficult to obtain a holistic view of a patient's health.
- **Information Overload:** The sheer volume of data can overwhelm healthcare providers, hindering their ability to identify critical patterns and trends.
- **Delayed Insights:** Traditional data processing methods can lead to delays in generating insights, impacting timely clinical decisions.
- **Lack of Predictive Capabilities:** Limited ability to proactively identify patients at risk or predict adverse events.

These issues can lead to suboptimal patient outcomes, increased healthcare costs, and physician burnout.

3. Solution Overview

DataDose addresses these challenges by providing a unified platform that:

- **Integrates Data:** Consolidates data from Electronic Health Records (EHRs), lab systems, and other sources into a central repository.
- **Builds a Knowledge Graph:** Utilizes Neo4j to model complex relationships between patients, diseases, treatments, and genetic factors, enabling sophisticated querying.
- **Enables Real-time Analytics:** Implements a streaming architecture for immediate data processing and insight generation.
- **Deploys AI Models:** Incorporates machine learning algorithms for predictive analytics and clinical decision support.
- **Provides Accessible Reporting:** Offers intuitive dashboards and reports through Power BI for easy consumption by clinicians and administrators.

The platform is designed with scalability, security, and ease of use in mind, ensuring it can be effectively deployed in diverse healthcare settings.

4. Project Objectives

The primary objectives of the DataDose project include:

- **Develop a robust data ingestion and processing pipeline:** Capable of handling large volumes of diverse healthcare data.
- **Implement a Neo4j knowledge graph:** To represent and query complex clinical relationships.
- **Establish a real-time data streaming capability:** For immediate data availability.
- **Build an analytics warehouse:** Optimized for clinical reporting and analysis.
- **Integrate AI/ML models:** For predictive patient risk stratification and disease progression.
- **Create a user-friendly API:** To facilitate integration with existing healthcare systems.
- **Design intuitive Power BI dashboards:** For visualizing key clinical metrics and insights.
- **Containerize the application using Docker:** For streamlined deployment and scalability.
- **Ensure data security and compliance:** Adhering to healthcare data privacy regulations (e.g., HIPAA).
- **Demonstrate improved clinical decision-making:** Through pilot testing and feedback.

5. System Architecture

The DataDose platform employs a layered, microservices-oriented architecture designed for flexibility and scalability. The core components include:

- **Data Sources:** EHRs, lab systems, medical devices.
- **Data Ingestion Layer:** Collects and preprocesses raw data.
- **Streaming Layer:** Processes real-time data using technologies like Kafka.
- **Data Storage:** Includes a Neo4j Knowledge Graph and an Analytics Warehouse (e.g., PostgreSQL).
- **Processing & Analytics Layer:** Houses ETL processes, Airflow for orchestration, and AI/ML models.
- **API Layer:** Exposes functionalities via RESTful APIs.
- **Presentation Layer:** Utilizes Power BI for reporting and dashboards.
- **Deployment Layer:** Docker and Kubernetes for containerization and orchestration.

This modular design allows for independent scaling and updates of different components.

6. Neo4j Knowledge Graph

The Neo4j knowledge graph is central to DataDose's ability to model and query complex relationships within healthcare data. It stores nodes representing entities such as:

- **Patients:** With attributes like demographics, medical history.
- **Diseases:** ICD codes, descriptions, symptoms.
- **Treatments:** Medications, procedures, therapies.
- **Doctors:** Specializations, affiliations.
- **Labs:** Test results, reference ranges.

Relationships (edges) define connections like , , , . This graph structure allows for powerful queries to uncover hidden correlations, identify at-risk patient cohorts, and understand treatment efficacy in ways traditional relational databases cannot.

Cypher Query Example: Find all patients diagnosed with diabetes who are prescribed a specific medication.

7. Data Pipeline

A robust data pipeline is essential for transforming raw data into actionable insights. The DataDose pipeline consists of the following stages:

1. **Extraction:** Data is extracted from various sources (EHRs, LIMS, etc.) using connectors or APIs.
2. **Transformation:** Data is cleaned, standardized, and validated. This includes handling missing values, normalizing units, and conforming to a common schema.
3. **Loading:** Transformed data is loaded into the Analytics Warehouse and potentially used to update the Neo4j Knowledge Graph.

This pipeline is designed to be **batch-oriented** for historical data and **stream-oriented** for real-time updates, ensuring data freshness and completeness.

8. Real-time Data Streaming

To provide up-to-the-minute insights, DataDose incorporates a real-time data streaming capability. Technologies like **Apache Kafka** are used as a distributed event streaming platform. Data events from critical systems are published to Kafka topics. Downstream consumers, including microservices responsible for updating the knowledge graph, triggering alerts, or feeding into the analytics warehouse, can subscribe to these topics.

This enables rapid detection of critical events (e.g., abnormal lab results, vital sign alerts) and ensures that the platform's insights reflect the most current patient status.

9. Analytics Warehouse

An optimized **Analytics Warehouse** serves as the central repository for structured and semi-structured data, supporting complex analytical queries and reporting. Technologies like **PostgreSQL** are leveraged for their reliability and powerful SQL capabilities.

The warehouse is designed with a dimensional modeling approach (e.g., star or snowflake schema) to facilitate efficient querying by business intelligence tools like Power BI. Key fact tables might include patient encounters, lab results, and medication administrations, with associated dimension tables for patients, doctors, dates, and locations.

10. Airflow Orchestration

Apache Airflow is used to programmatically author, schedule, and monitor workflows. In DataDose, Airflow manages the complex dependencies within the data pipeline.

Key functionalities managed by Airflow include:

- Scheduling batch ETL jobs for data loading and transformation.
- Orchestrating the synchronization between the Analytics Warehouse and the Neo4j Knowledge Graph.
- Triggering AI model training and retraining processes.
- Managing data quality checks and alerting on failures.

Airflow's DAG (Directed Acyclic Graph) system provides a clear, visual representation of workflows, making it easier to manage and troubleshoot the entire data processing lifecycle.

11. AI Clinical Features

DataDose integrates several **Artificial Intelligence (AI)** and **Machine Learning (ML)** features to enhance clinical decision-making:

- **Patient Risk Stratification:** Models predict the likelihood of a patient developing specific conditions (e.g., sepsis, heart failure) or experiencing adverse events based on their historical and real-time data.
- **Disease Progression Prediction:** Algorithms forecast the likely progression of chronic diseases, allowing for proactive interventions.
- **Treatment Recommendation Support:** AI can suggest potential treatment pathways based on similar patient profiles and treatment outcomes, aiding clinicians in personalized medicine.
- **Anomaly Detection:** Identifying unusual patterns in patient vitals or lab results that might indicate a deteriorating condition.

These AI features are developed using Python libraries (e.g., Scikit-learn, TensorFlow) and integrated into the platform via the API layer.

12. Application API

A well-defined **Application Programming Interface (API)** serves as the gateway for external systems and internal microservices to interact with DataDose. The API is built using a RESTful architecture, ensuring broad compatibility.

Key API endpoints allow for:

- **Data Retrieval:** Fetching patient summaries, lab results, risk scores, and knowledge graph insights.
- **Data Ingestion:** Programmatically submitting new data points or patient records.
- **Alerting:** Receiving real-time notifications for critical events.

- **Integration:** Enabling seamless connection with existing EHR systems or other healthcare applications.

The API is secured using standard authentication and authorization mechanisms to protect sensitive patient data.

13. Technology Stack

The DataDose platform is built using a modern and robust technology stack:

- **Programming Languages:** Python (Backend, AI/ML), JavaScript (Frontend - if applicable).
- **Databases:** Neo4j (Knowledge Graph), PostgreSQL (Analytics Warehouse).
- **Streaming:** Apache Kafka.
- **Orchestration:** Apache Airflow.
- **Containerization:** Docker.
- **Cloud/Deployment:** Potentially AWS, Azure, GCP or on-premise servers.
- **BI/Reporting:** Microsoft Power BI.
- **API Framework:** FastAPI (Python).
- **AI/ML Libraries:** Scikit-learn, TensorFlow, PyTorch.

This stack provides a balance of performance, scalability, and developer productivity.

14. Power BI Reporting

Microsoft Power BI is utilized to create interactive and insightful reports and dashboards for clinicians and administrators. These visualizations provide a user-friendly interface to explore the data processed by DataDose.

Key Dashboard Examples:

- **Patient Dashboard:** Overview of a single patient's key metrics, recent events, and risk scores.
- **Population Health Dashboard:** Trends in disease prevalence, treatment adherence across patient cohorts.
- **Operational Dashboard:** Monitoring system performance, data pipeline status, and resource utilization.

Power BI connects directly to the Analytics Warehouse, enabling real-time or near real-time updates of the reports.

15. Docker Deployment

To ensure consistent, reproducible, and scalable deployments, DataDose leverages **Docker** containerization. Each microservice and core component (e.g., Neo4j, Kafka, Airflow) is packaged into a Docker image.

Benefits of Docker Deployment:

- **Environment Consistency:** Eliminates 'it works on my machine' issues.
- **Scalability:** Easily scale services up or down by launching more containers.
- **Isolation:** Services run in isolated environments, preventing conflicts.
- **Simplified Management:** Streamlines installation, updates, and rollbacks.

For larger deployments, **Docker Compose** or **Kubernetes** can be used for orchestration and management of multiple containers.

16. Key Performance Indicators (KPIs)

The success of the DataDose platform is measured against several Key Performance Indicators (KPIs):

- **Reduction in Average Patient Length of Stay:** Indicating more efficient care management.
- **Decrease in Readmission Rates:** Demonstrating improved post-discharge care and monitoring.
- **Improvement in Time-to-Diagnosis/Treatment:** Reflecting faster identification and intervention.
- **Increase in Clinician Satisfaction:** Measured through user feedback on the platform's utility.
- **Accuracy of AI Predictions:** Measured by standard ML metrics (e.g., AUC, Precision, Recall).
- **Data Latency:** Time taken from data generation to its availability in the platform.
- **System Uptime:** Ensuring continuous availability of the platform.

17. Metrics and Monitoring

Comprehensive monitoring and metrics collection are vital for maintaining the health and performance of DataDose.

Key Metrics Tracked:

- **System Performance:** CPU usage, memory consumption, network traffic for all services.
- **Data Pipeline Throughput:** Volume of data processed per unit time.
- **Error Rates:** Frequency and types of errors encountered in data processing and API calls.
- **Database Performance:** Query response times, connection pool usage for Neo4j and PostgreSQL.
- **Kafka Consumer Lag:** Ensuring real-time data streams are being processed effectively.
- **API Latency and Error Rates:** Monitoring the responsiveness and reliability of the API.

Tools like Prometheus and Grafana can be integrated for real-time monitoring and alerting.

18. Team and Roles

The DataDose project was a collaborative effort involving a dedicated team:

- **Project Manager:** Overseeing project timelines, resources, and communication.
- **Lead Architect:** Designing the overall system architecture and technology choices.
- **Data Engineers:** Building and maintaining the data pipelines, ETL processes, and data warehouse.
- **Knowledge Graph Specialist:** Designing, implementing, and querying the Neo4j graph database.
- **AI/ML Engineers:** Developing, training, and deploying predictive models.
- **Backend Developers:** Building the core application logic and APIs.
- **Frontend/Bi Developers:** Creating user interfaces and Power BI reports/dashboards.
- **DevOps Engineer:** Managing containerization, deployment, and infrastructure.

Each member brought specialized skills contributing to the project's success.

19. Challenges and Lessons Learned

Throughout the development of DataDose, the team encountered several challenges:

- **Data Quality and Standardization:** Inconsistent data formats and missing values from source systems required significant cleaning efforts.
- **Integration Complexity:** Integrating diverse systems and technologies posed technical hurdles.
- **Scalability Testing:** Ensuring the platform could handle production-level data volumes required rigorous testing.
- **Real-time Processing Latency:** Optimizing the streaming pipeline to minimize data latency was a continuous effort.
- **Domain Expertise:** Understanding complex clinical workflows and terminology required close collaboration with healthcare professionals.

Key Lessons Learned:

- **Iterative Development:** Adopting an agile approach allowed for flexibility and continuous improvement.
- **Early and Frequent Testing:** Identifying and resolving issues early in the development cycle.
- **Clear Documentation:** Maintaining detailed documentation for architecture, pipelines, and APIs is crucial for collaboration and maintenance.
- **Cross-functional Collaboration:** Effective communication between data engineers, AI specialists, and domain experts is paramount.

20. Conclusion

DataDose represents a significant advancement in Clinical Decision Intelligence, providing healthcare providers with the tools to make more informed, timely, and personalized patient care decisions. The platform's sophisticated architecture, leveraging a Neo4j knowledge graph, real-time streaming, AI-driven insights, and user-friendly reporting, addresses critical challenges in modern healthcare data management.

This graduation project successfully demonstrates the potential of integrated data intelligence to improve patient outcomes, enhance operational efficiency, and support evidence-based medical practice. The lessons learned and technologies employed lay a strong foundation for future development and broader adoption of Clinical Decision Intelligence platforms.